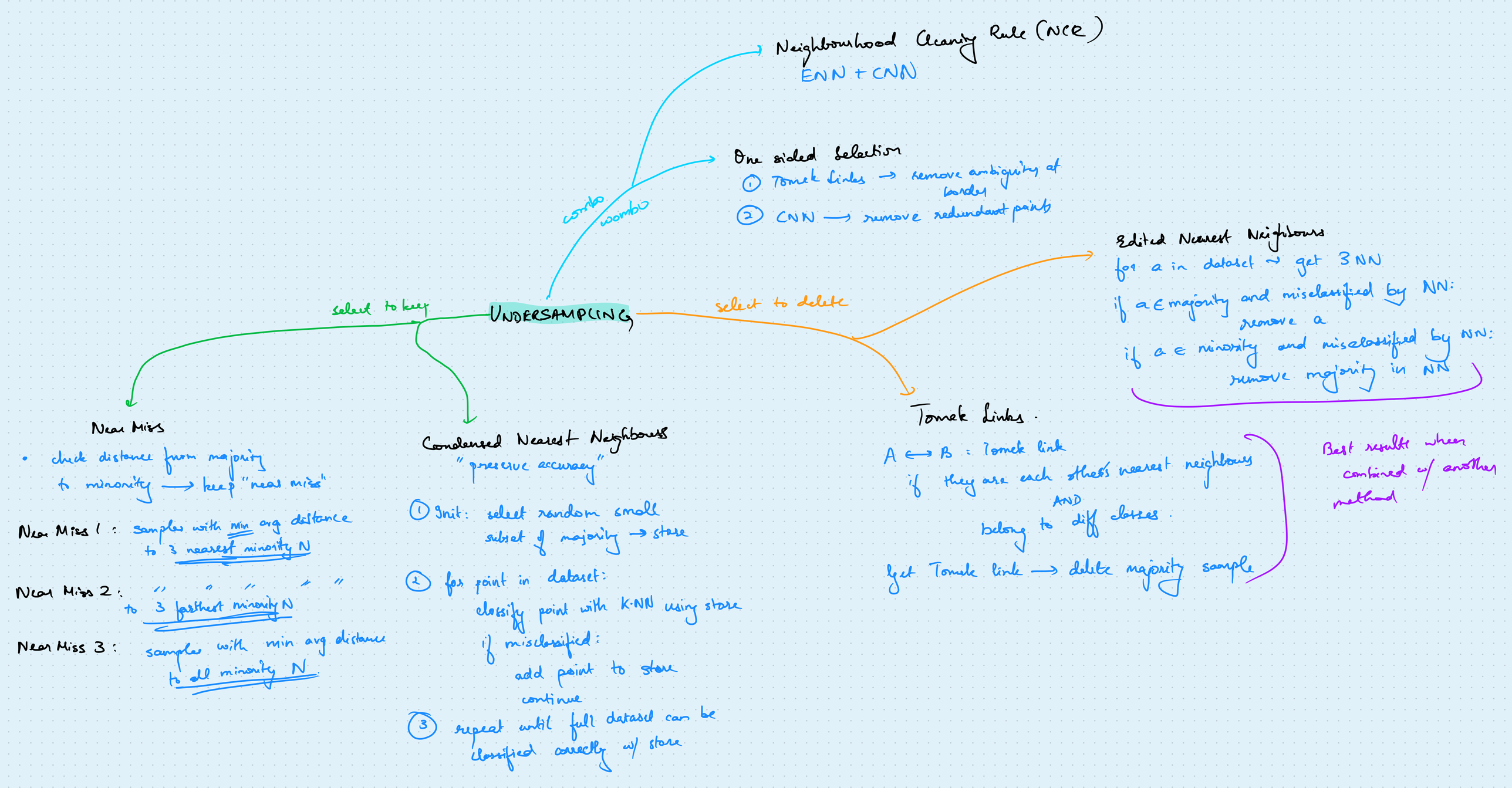

UNDERSAMPLING
select to keep
select to delete
combo
combo
Neighbourhood Cleaning Rule (NCR)
ENN + CNN
One sided selection
1 Tomek links → remove ambiguity at border
2 CNN → remove redundant points
Edited Nearest Neighbours
for a in dataset → get 3 NN
if a ∈ majority and misclassified by NN: remove a
if a ∈ minority and misclassified by NN: remove majority in NN
Near Miss
• check distance from majority to minority → keep "near miss"
Near Miss 1 : samples with min avg distance to 3 nearest minority N
Near Miss 2 : " " " " " to 3 farthest minority N
Near Miss 3 : samples with min avg distance to all minority N
Condensed Nearest Neighbours
"preserve accuracy"
1 Init: select random small subset of majority → store
2 for point in dataset: classify point with k-NN using store
if misclassified: add point to store
continue
3 repeat until full dataset can be classified correctly w/ store
Tomek Links
A ⟷ B : Tomek link
if they are each others nearest neighbours AND belong to diff classes.
get Tomek link → delete majority sample
Best results when combined w/ another method